

MVC-Assignment-2

Sana Munir Alam
24K-0573
BSCS-2E

$$Q1) C(x, y) = e^x + \sin(y) - 4x - y$$

$$f_x = e^x - 4$$

$$f_y = \cos(y) - 1$$

$$f_{xy} = 0$$

$$f_{xx} = e^x$$

$$f_{yy} = -\sin(y)$$

$$f_{xx} = 0$$

$$e^x - 4 = 0$$

$$e^x = 4$$

$$x = \ln 4$$

$$f_y = 0$$

$$\cos(y) - 1 = 0$$

$$y = \cos^{-1}(1)$$

$$y = 0$$

$\therefore (x, y) \rightarrow (\ln 4, 0)$
is critical point

$$f_{xx}(\ln 4, 0) = e^{\ln 4} = 4$$

$$f_{yy}(\ln 4, 0) = -\sin(0) = 0$$

$$f_{xy}(\ln 4, 0) = 0$$

$$\therefore D = f_{xx}f_{yy} - (f_{xy})^2$$

$$D = 4(0) - 0^2$$

$$D = 0$$

as $D = 0$ the test fails

$$Q2) M(a, b) = 100a + 150b - 2a^2 - 3b^2 - ab$$

$$f_a = 100 - 4a - b$$

$$f_b = 150 - 6b - a$$

$$f_{ab} = 0$$

$$f_{aa} = -4$$

$$f_{bb} = -6$$

$$f_a = 0$$

$$100 - 4a - b = 0$$

$$b = 100 - 4a \rightarrow \text{eq 1}$$

$$f_b = 0$$

$$150 - 6b - a = 0 \rightarrow \text{eq 2}$$

substitute eq 1 in eq 2

$$150 - 6(100 - 4a) - a = 0$$

$$150 - 600 + 24a - a = 0$$

$$23a = 450$$

$$a = \frac{450}{23}$$

for b

$$150 - 6b - \frac{450}{23} = 0$$

$$b = \frac{500}{23}$$

$$\therefore \left(\frac{a}{2}, \frac{b}{2} \right) \rightarrow \left(\frac{450}{23}, \frac{500}{23} \right) \text{ critical point}$$

$$f_{aa} \left(\frac{450}{23}, \frac{500}{23} \right) = -4$$

$$f_{bb} \left(\frac{450}{23}, \frac{500}{23} \right) = -6$$

$$f_{ab} \left(\frac{450}{23}, \frac{500}{23} \right) = 0$$

$$\begin{aligned} \therefore D &= f_{aa} f_{bb} - (f_{ab})^2 \\ &= -4(-6) - (0)^2 \\ &= 24 \end{aligned}$$

as $D > 0$ and $f_{aa} < 0 \therefore f$ has relative maximum at $\left(\frac{450}{23}, \frac{500}{23} \right)$

Q3) $C(m, n) = 3m^2 + 2mn + 4n^2$ } function

$g(m, n) = m^2 + n^2 + mn - 400$ } constraint

$$C_m = \lambda g_m$$

$$6m + 2n = \lambda (2m + n)$$

$$\lambda = \frac{6m + 2n}{2m + n}$$

$$C_n = \lambda g_n$$

$$8n + 2m = \lambda (2n + m)$$

$$\lambda = \frac{2m + 8n}{m + 2n}$$

compare

$$\frac{6m + 2n}{2m + n} = \frac{2m + 8n}{m + 2n}$$

$$(6m + 2n)(m + 2n) = (2m + 8n)(2m + n)$$

$$6m^2 + 12mn + 2mn + 4n^2 = 4m^2 + 2mn + 16mn + 8n^2$$

$$6m^2 - 4m^2 + 14mn - 18mn + 4n^2 - 8n^2 = 0$$

$$2m^2 - 4mn - 4n^2 = 0$$

$$m^2 - 2mn - 2n^2 = 0$$

$$m^2 = 2mn + 2n^2 \rightarrow \text{eq 1}$$

put eq 1 in constraint

$$m^2 + n^2 + mn - 400 = 0$$

$$\cancel{m^2} = \cancel{400} \cancel{m^2}$$

$$n^2 + mn = 400 - m^2$$

$$\frac{m^2}{2} = 800 - m^2$$

$$m^2 = 800 - 2m^2$$

$$3m^2 = 800$$

$$m = \pm \sqrt{\frac{800}{3}} \approx \pm 16.329$$

$$\text{when } m = + \sqrt{\frac{800}{3}}$$

$$\left(\sqrt{\frac{800}{3}}\right)^2 + n^2 + n\left(\sqrt{\frac{800}{3}}\right) = 400$$

$$n^2 + \frac{20\sqrt{2}}{3}n - \frac{400}{3} = 0$$

$$n = 5.977 \text{ and } n = -22.30$$

$$\text{when } m = - \sqrt{\frac{800}{3}}$$

$$\left(-\sqrt{\frac{800}{3}}\right)^2 + n^2 + n\left(-\sqrt{\frac{800}{3}}\right) = 400$$

$$n^2 - \frac{20\sqrt{2}}{3}n + \frac{800}{3} = 400$$

$$n = 22.3 \text{ and } n = -5.977$$

$$\therefore (16.329, 5.977) \quad (-16.329, 22.30) \\ (16.329, -22.30) \quad (-16.329, -5.977)$$

$$\textcircled{Q4) } E(p, q, r) = p^2 + 4q^2 + 2r^2 \quad \left. \begin{array}{l} \text{\{ function} \end{array} \right\}$$

$$g(p, q, r) = p^2 + q^2 + r^2 = 250000 \quad \left. \begin{array}{l} \text{\{ constraint} \end{array} \right\}$$

$$E_p = \lambda g_p \quad \left| \quad E_q = \lambda g_q \quad \left| \quad E_r = \lambda g_r \right.$$

$$2p = \lambda 2p \quad \left| \quad 8q = \lambda 2q \quad \left| \quad 4r = \lambda 2r \right.$$

$$\lambda = 1 \quad \left| \quad \lambda = 4 \quad \left| \quad \lambda = 2 \right.$$

putting all energy in p so $q=0$ and $r=0$

$$p^2 + q^2 + r^2 = 250000$$

$$\sqrt{p^2} = \sqrt{250000}$$

$$p = \pm 500$$

putting all energy in q so $p=0$ and $r=0$

$$p^2 + q^2 + r^2 = 250000$$

$$\sqrt{q^2} = \sqrt{250000}$$

$$q = \pm 500$$

putting all energy in r so $p=0$ and $q=0$

$$p^2 + q^2 + r^2 = 250,000$$

$$\sqrt{r^2} = \sqrt{250000}$$

$$r = \pm 500$$

$$\therefore (0, 0, \pm 500), (0, \pm 500, 0), (\pm 500, 0, 0)$$

Putting these values in functions

$$E(0, 0, \pm 500) = 500000$$

$$E(0, \pm 500, 0) = 1000000$$

$$E(\pm 500, 0, 0) = 250,000 \quad \text{\{ most minimum value we get}$$

As we needed to minimize E , we spent all our energy in the direction that increase E the least. In this case we will put all our energy in p , which gives 250000 in answer which is the most minimum energy consumption

$$Q5) J(\theta_1, \theta_2) = \sin(\theta_1) + \theta_2^2 + \theta_1 \theta_2$$

initial value (1,1) and learning rate $\alpha = 0.1$

$$J_{\theta_1} = \cos(\theta_1) + \theta_2$$

using radian
mode

$$J_{\theta_2} = 2\theta_2 + \theta_1$$

put $\theta_1 = 1$ and $\theta_2 = 1$ in J_{θ_1} and J_{θ_2}

$$J_{\theta_1} = \cos(1) + 1 = 1.54$$

$$J_{\theta_2} = 2(1) + 1 = 3$$

first iteration answer :- $J(\theta_1, \theta_2) = J(1.54, 3)$

$$\left. \begin{aligned} \theta_1^{\text{new}} &= \theta_1^{\text{old}} - \alpha J_{\theta_1} \\ \theta_1^{\text{new}} &= 1 - 0.1(1.54) \\ \theta_1^{\text{new}} &= 0.84597 \end{aligned} \right| \begin{aligned} \theta_2^{\text{new}} &= \theta_2^{\text{old}} - \alpha J_{\theta_2} \\ &= 3 - 0.1(3) \\ &= 2.7 \end{aligned}$$

second iteration answer

$$\left. \begin{aligned} \theta_1^{\text{new}} &= \theta_1^{\text{old}} - \alpha J_{\theta_1} \\ \theta_1^{\text{new}} &= 0.84597 - 0.1(1.54) \\ \theta_1^{\text{new}} &= \end{aligned} \right\} \begin{aligned} J(\theta_1, \theta_2) &= (0.84597, 0.7) \\ \text{put in } J(\theta_1, \theta_2) \text{ in equation} \\ &= \sin(0.84597) + 0.7^2 + (0.84597)(0.7) \\ &= 1.8307 \end{aligned}$$

for second iteration

$$J_{\theta_1} = \cos(0.84597) + 0.7 = 1.363$$

$$J_{\theta_2} = 2(0.7) + 0.84597 = 2.24597$$

$$\left. \begin{aligned} \theta_1^{\text{new}} &= \theta_1^{\text{old}} - \alpha J_{\theta_1} \\ \theta_1^{\text{new}} &= 0.84597 - 0.1(1.363) \\ &= 0.70967 \end{aligned} \right| \begin{aligned} \theta_2^{\text{new}} &= \theta_2^{\text{old}} - \alpha J_{\theta_2} \\ &= 0.7 - 0.1(2.24597) \\ &= 0.4753 \end{aligned}$$

$$(\theta_1, \theta_2) = (0.70967, 0.4753)$$

put in $J(\theta_1, \theta_2)$ value in equation

$$J(0.70967, 0.4753) = \sin(0.70967) + 0.4753^2 + (0.70967)(0.4753) \\ = 1.2147$$

∴ first iteration value : 1.8307

Second iteration value : 1.2147

Q6) $R(f, g, h) = f^2 + 2g^2 + 3h^2 + fg - 5f - 7g - 9h + 30$

initial value $(1, 1, 1)$ and rate $\alpha = 0.05$

$$R_f = 2f + g - 5$$

$$\text{put } R_f(1, 1, 1) = 2 + 1 - 5 = -2$$

$$R_g = 4g + f - 7$$

$$R_g(1, 1, 1) = 4 + 1 - 7 = -2$$

$$R_h = 6h - 9$$

$$R_h(1, 1, 1) = 6 - 9 = -3$$

first iteration

$$f^{\text{new}} = 1 - 0.05(-2) \\ = 1.1$$

$$g^{\text{new}} = 1 - 0.05(-2) \\ = 1.1$$

$$h^{\text{new}} = 1 - 0.05(-3) \\ = 1.15$$

$$(f, g, h) = (1.1, 1.1, 1.15)$$

put in $R(f, g, h) = 15.2575$

$$R_f(1.1, 1.1, 1.15) = -1.7$$

$$R_g(1.1, 1.1, 1.15) = -1.5$$

$$R_h(1.1, 1.1, 1.15) = -2.1$$

second iteration

$$f^{\text{new}} = 1.1 - 0.05(-1.7) \quad \left| \quad g^{\text{new}} = 1.1 - 0.05(-1.5) \quad \left| \quad h^{\text{new}} = 1.15 - 0.05(-2.1) \right. \right. \\ = 1.185 \quad \left. \quad \quad \quad = 1.175 \quad \quad \quad = 1.255$$

$$(f, g, h) = (1.185, 1.175, 1.255)$$

put in $R(f, g, h) = 14.8379$

$$R_f (1.185, 1.175, 1.255) = -1.455$$

$$R_g (1.185, 1.175, 1.255) = -1.115$$

$$R_h (1.185, 1.175, 1.255) = -1.47$$

third iteration

$$\begin{array}{l|l|l} f^{\text{new}} = 1.185 - 0.05(-1.455) & g^{\text{new}} = 1.175 - 0.05(-1.115) & h^{\text{new}} = 1.255 - 0.05(-1.47) \\ = 1.25775 & = 1.23075 & = 1.3285 \end{array}$$

$$(f, g, h) = (1.25775, 1.23075, 1.3285)$$

put in $R(f, g, h) = 14.593$

$$R_f (1.25775, 1.23075, 1.3285) = -1.25375$$

$$R_g (1.25775, 1.23075, 1.3285) = -0.81925$$

$$R_h (1.25775, 1.23075, 1.3285) = -1.029$$

fourth iteration

$$\begin{array}{l|l} f^{\text{new}} = 1.25775 - 0.05(-1.25375) & g^{\text{new}} = 1.23075 - 0.05(-0.81925) \\ = 1.32043 & = 1.27171 \end{array}$$

$$\begin{aligned} h^{\text{new}} &= 1.3285 - 0.05(-1.029) \\ &= 1.37995 \end{aligned}$$

$$(f, g, h) = (1.32043, 1.27171, 1.37995)$$

put in $R(f, g, h) = 14.4463$

$$\therefore \text{zero iteration} = \cancel{15.257}$$

$$\text{first iteration} = 15.257$$

$$\text{second iteration} = 14.837$$

$$\text{third iteration} = 14.593$$

$$\text{fourth iteration} = 14.446$$

$$Q7) J(\theta_1, \theta_2, \theta_3) = e^{\theta_1^2} + \theta_2^2 + 4\theta_3^2 + \theta_1\theta_2$$

find the optimal value using the Hessian matrix

$$J_{\theta_1} \Rightarrow 2\theta_1 e^{\theta_1^2} + \theta_2 = 0 \quad \rightarrow \text{eq 1}$$

$$J_{\theta_2} \Rightarrow 2\theta_2 + \theta_1 = 0$$

$$J_{\theta_3} \Rightarrow 8\theta_3 = 0 \quad \therefore J_{\theta_3} \text{ is } \theta_3 = 0$$

for θ_2 and θ_1

$$2\theta_2 + \theta_1 = 0$$

$$\theta_2 = -\frac{\theta_1}{2} \rightarrow \text{eq 2}$$

substitute eq 2 in eq 1

$$2\theta_1 e^{\theta_1^2} - \frac{\theta_1}{2} = 0$$

$$4\theta_1 e^{\theta_1^2} - \theta_1 = 0$$

$$\theta_1 (4e^{\theta_1^2} - 1) = 0$$

$$\theta_1 = 0 \quad \text{and} \quad e^{\theta_1^2} = \frac{1}{4}$$

no real solution

for θ_2

$$\theta_2 = -\frac{\theta_1}{2}$$

$$\theta_2 = -\frac{0}{2}$$

$$\theta_2 = 0$$

$$\therefore \theta_1 = 0 \text{ and } \theta_2 = 0 \text{ and } \theta_3 = 0$$

Now for Hessian matrix

$$H = \begin{bmatrix} \frac{\partial^2 J}{\partial \theta_1^2} & \frac{\partial^2 J}{\partial \theta_1 \partial \theta_2} & \frac{\partial^2 J}{\partial \theta_1 \partial \theta_3} \\ \frac{\partial^2 J}{\partial \theta_2 \partial \theta_1} & \frac{\partial^2 J}{\partial \theta_2^2} & \frac{\partial^2 J}{\partial \theta_2 \partial \theta_3} \\ \frac{\partial^2 J}{\partial \theta_3 \partial \theta_1} & \frac{\partial^2 J}{\partial \theta_3 \partial \theta_2} & \frac{\partial^2 J}{\partial \theta_3^2} \end{bmatrix}$$

finding second derivative

$$J_{\theta_1 \theta_1} = 4\theta_1^2 e^{\theta_1^2} + 2e^{\theta_1^2} \quad \text{at } (0,0,0) = J_{\theta \theta_1} = 2$$

$$\begin{array}{c|c|c} J_{\theta_1 \theta_2} = 1 & J_{\theta_2 \theta_2} = 2 & J_{\theta_3 \theta_2} = 0 \\ J_{\theta_1 \theta_3} = 0 & J_{\theta_2 \theta_3} = 0 & J_{\theta_3 \theta_3} = 8 \\ J_{\theta_2 \theta_1} = 1 & J_{\theta_3 \theta_1} = 0 & \end{array}$$

$$H = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 8 \end{bmatrix}$$

$$D_1 = |2| = 2$$

$$D_2 = \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} = 4 - 1 = 3$$

$$D_3 = \begin{vmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 8 \end{vmatrix} = 2(8(2) - 0) - 1(1(8) - 0) + 0(1(0) - 2(0))$$

$$= 2(16) - 8$$

$$= 32 - 8$$

$$= 24$$

as all d are positive so it's positive definite which means it's a convex and it's minimum value at $(0,0,0)$

Q8) $T(x, y) = xy - x^2$

region bounded by

$$0 \leq x \leq 1$$

$$1 \leq y \leq 3$$

$$E = \iint_{[0]} xy - x^2 dA$$

$$E = \int_1^3 \int_0^1 xy - x^2 dx dy$$

$$E = \int_1^3 \left[\int_0^1 xy - x^2 dx \right] dy$$

$$E = \int_1^3 \left[\frac{x^2 y}{2} - \frac{x^3}{3} \right]_0^1 dy$$

$$E = \int_1^3 \left[\left(\frac{y}{2} - \frac{1}{3} \right) - \left(\frac{0}{2} - \frac{0}{3} \right) \right] dy$$

$$E = \int_1^3 \frac{y}{2} - \frac{1}{3} dy$$

$$E = \left[\frac{y^2}{4} - \frac{y}{3} \right]_1^3$$

$$E = \left[\frac{9}{4} - \frac{3}{3} \right] - \left[\frac{1}{4} - \frac{1}{3} \right]$$

$$E = \frac{4}{3} \text{ Ans}$$

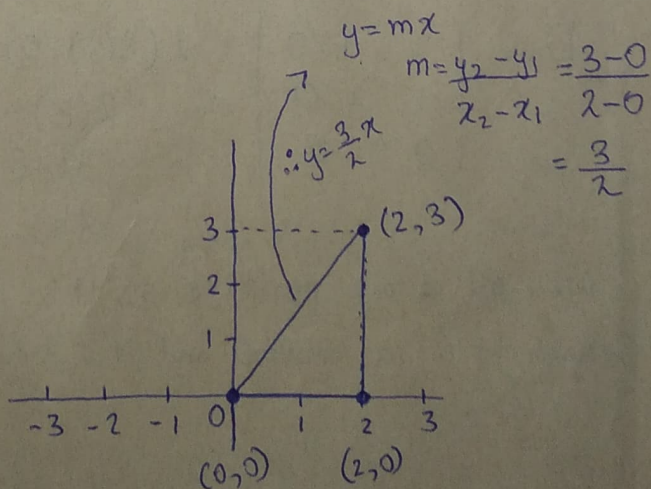
Q9) $z = 10 - x^2 - y^2$

Data :-

vertices at (0,0)

(2,0)

(2,3)



$$V = \int_0^2 \int_0^{\frac{3x}{2}} (10 - x^2 - y^2) dy dx$$

$$V = \int_0^2 \left[\int_0^{\frac{3x}{2}} (10 - x^2 - y^2) dy \right] dx$$

$$V = \int_0^2 \left[10y - x^2y - \frac{y^3}{3} \right]_0^{\frac{3x}{2}} dx$$

$$V = \int_0^2 \left[\left(10 \left[\frac{3x}{2} \right] - x^2 \left[\frac{3x}{2} \right] - \frac{\left(\frac{3x}{2} \right)^3}{3} \right) - (0 - 0 - 0) \right] dx$$

$$V = \int_0^2 \left[\frac{30x}{2} - \frac{3x^3}{2} - \frac{27x^3}{8} \times \frac{1}{3} \right] dx$$

$$V = \int_0^2 \frac{30x}{2} - \frac{3x^3}{2} - \frac{9x^3}{8} dx$$

$$V = \left[\frac{30x^2}{4} - \frac{3x^4}{8} - \frac{9x^4}{32} \right]_0^2$$

$$V = \left[\frac{30(2)^2}{4} - \frac{3(2)^4}{8} - \frac{9(2)^4}{32} \right] - (0 - 0 - 0)$$

$$V = 30 - 6 - \frac{9}{2}$$

$$\boxed{V = 19.5 \text{ cubic units}} \text{ Ans}$$

Q10a) $f(x, y, z) = x^2 + y^2 - z = 0$ at point $(1, 2, 5)$

$$f_x = 2x \quad \text{at } f_x(1, 2, 5) = 2$$

$$f_y = 2y \quad \text{at } f_y(1, 2, 5) = 4$$

$$f_z = -1 \quad \text{at } f_z(1, 2, 5) = -1$$

equation of tangent plane

$$f_x(x-x_0) + f_y(y-y_0) + f_z(z-z_0) = 0$$

$$2(x-1) + 4(y-2) - 1(z-5) = 0$$

$$2x - 2 + 4y - 8 - z + 5 = 0$$

$$2x + 4y - z - 5 = 0$$

equation of normal plane

$$x = x_0 + f_x t \quad \left| \quad y = y_0 + f_y t \quad \left| \quad z = z_0 + f_z t \right. \right.$$

$$x = 1 + 2t \quad \left| \quad y = 2 + 4t \quad \left| \quad z = 5 - t \right. \right.$$

Q10b) $f_1(x, y, z) = x^2 + y^2 + z^2 - 9 = 0$

$$f_2(x, y, z) = x + 2y - 3z + 4 = 0$$

find angle between their tangent plane at intersection point $(2, -1, 1)$.

For surface 1

$$f_x = 2x \quad \text{at } f_x(2, -1, 1) = 4$$

$$f_y = 2y \quad \text{at } f_y(2, -1, 1) = -2$$

$$f_z = 2z \quad \text{at } f_z(2, -1, 1) = -2$$

$$\nabla f_1 = 4i - 2j + 2k$$

for surface 2

$$f_x = 1 \quad \text{at } f_x(2, -1, 1) = 1$$

$$f_y = 2 \quad \text{at } f_y(2, -1, 1) = 2$$

$$f_z = -3 \quad \text{at } f_z(2, -1, 1) = -3$$

$$\nabla f_2 = i + 2j - 3k$$

$$\cos \theta = \frac{\nabla f_1 \cdot \nabla f_2}{|\nabla f_1| |\nabla f_2|}$$

$$|\nabla f_1| = \sqrt{4^2 + (-2)^2 + 2^2} = 2\sqrt{6}$$

$$|\nabla f_2| = \sqrt{1^2 + 2^2 + (-3)^2} = \sqrt{14}$$

$$\theta = \cos^{-1} \left[\frac{4 - 4 - 6}{(2\sqrt{6})(\sqrt{14})} \right]$$

$$\theta = 1.904 \text{ radians}$$